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Task 1 – Mega-menu (HTML + CSS only)
index.html – hamburger, nav, "Shop by Category" mega-menu (lines 29–143)
The identical block is also in product.html (lines 27–141) and checkout.html (lines 27–141).

```
    <!-- CSS-only hamburger: the checkbox holds the open/closed state (no JavaScript) -->
    <input type="checkbox" id="menu-toggle" class="menu-toggle" aria-label="Open menu">
    <label for="menu-toggle" id="hamburger">
      <svg viewBox="0 0 24 24" fill="none" stroke="white" stroke-width="2"
stroke-linecap="round">
        <line x1="3" y1="6" x2="21" y2="6" />
        <line x1="3" y1="12" x2="21" y2="12" />
        <line x1="3" y1="18" x2="21" y2="18" />
      </svg>
    </label>
    <img src="images/arsenal-cannon.svg" id="arsenal-cannon">
    <nav aria-label="Main navigation">
      <div class="nav-menu">
        <!-- Task 1: "Shop by Category" mega-menu. Number of columns = K = 4 -->
        <div class="mega-item">
          <a href="#" class="nav-link nav-link-stack mega-trigger" aria-haspopup="true">SHOP
BY <br>CATEGORY</a>

          <!-- BP (643px) and wider: full-width panel, opens on :hover / :focus-within -->
          <div class="mega-panel">
          <div class="mega-col">
            <h2 class="mega-heading">Kit</h2>
            <ul>
              <li><a href="#" class="mega-link">Home Kit</a></li>
              <li><a href="#" class="mega-link">Away Kit</a></li>
              <li><a href="#" class="mega-link">Third Kit</a></li>
              <li><a href="#" class="mega-link">Goalkeeper Kit</a></li>
              <li><a href="#" class="mega-link">Training Wear</a></li>
            </ul>
          </div>
          <div class="mega-col">
            <h2 class="mega-heading">Clothing</h2>
            <ul>
              <li><a href="#" class="mega-link">T-Shirts</a></li>
              <li><a href="#" class="mega-link">Hoodies &amp;amp; Sweatshirts</a></li>
              <li><a href="#" class="mega-link">Jackets &amp;amp; Coats</a></li>
              <li><a href="#" class="mega-link">Polo Shirts</a></li>
```

```
    <li><a href="#" class="mega-link">Shorts</a></li>
  </ul>
</div>
<div class="mega-col">
  <h2 class="mega-heading">Accessories</h2>
  <ul>
    <li><a href="#" class="mega-link">Scarves &amp;amp; Hats</a></li>
    <li><a href="#" class="mega-link">Bags &amp;amp; Backpacks</a></li>
    <li><a href="#" class="mega-link">Mugs &amp;amp; Homeware</a></li>
    <li><a href="#" class="mega-link">Phone Cases</a></li>
    <li><a href="#" class="mega-link">Stationery</a></li>
  </ul>
</div>
<div class="mega-col">
  <h2 class="mega-heading">Shop For</h2>
  <ul>
    <li><a href="#" class="mega-link">Men</a></li>
    <li><a href="#" class="mega-link">Women</a></li>
    <li><a href="#" class="mega-link">Kids</a></li>
    <li><a href="#" class="mega-link">Baby &amp;amp; Toddler</a></li>
    <li><a href="#" class="mega-link">Retro Shirts</a></li>
  </ul>
</div>
</div>

<!-- Below BP: same groups as an accordion inside the hamburger menu -->
<div class="mega-accordion">
  <p class="mega-acc-title">SHOP BY CATEGORY</p>
<details>
  <summary>Kit</summary>
  <ul>
    <li><a href="#" class="mega-link">Home Kit</a></li>
    <li><a href="#" class="mega-link">Away Kit</a></li>
    <li><a href="#" class="mega-link">Third Kit</a></li>
    <li><a href="#" class="mega-link">Goalkeeper Kit</a></li>
    <li><a href="#" class="mega-link">Training Wear</a></li>
  </ul>
</details>
<details>
  <summary>Clothing</summary>
```

```
      <ul>
        <li><a href="#" class="mega-link">T-Shirts</a></li>
        <li><a href="#" class="mega-link">Hoodies &amp;amp; Sweatshirts</a></li>
        <li><a href="#" class="mega-link">Jackets &amp;amp; Coats</a></li>
        <li><a href="#" class="mega-link">Polo Shirts</a></li>
        <li><a href="#" class="mega-link">Shorts</a></li>
      </ul>
    </details>
    <details>
      <summary>Accessories</summary>
      <ul>
        <li><a href="#" class="mega-link">Scarves &amp;amp; Hats</a></li>
        <li><a href="#" class="mega-link">Bags &amp;amp; Backpacks</a></li>
        <li><a href="#" class="mega-link">Mugs &amp;amp; Homeware</a></li>
        <li><a href="#" class="mega-link">Phone Cases</a></li>
        <li><a href="#" class="mega-link">Stationery</a></li>
      </ul>
    </details>
    <details>
      <summary>Shop For</summary>
      <ul>
        <li><a href="#" class="mega-link">Men</a></li>
        <li><a href="#" class="mega-link">Women</a></li>
        <li><a href="#" class="mega-link">Kids</a></li>
        <li><a href="#" class="mega-link">Baby &amp;amp; Toddler</a></li>
        <li><a href="#" class="mega-link">Retro Shirts</a></li>
      </ul>
    </details>
    </div>
  </div>

  <a href="#" class="nav-link nav-link-stack">NEW <br>IN</a>
  <a href="#" class="nav-link">KIT</a>
  <a href="#" class="nav-link">ADIDAS</a>
  <a href="#" class="nav-link">CL0THING</a>
  <a href="#" class="nav-link">ACCESSORIES</a>
  <a href="#" class="nav-link">CHAMPIONS</a>
  <a href="#" class="nav-link">GIFTS</a>
  <a href="#" class="nav-link nav-link-stack">STADIUM <br>TOURS</a>
</div>
```

style.css – baseline rules that I changed
header z-index raised to 2000; position: relative added; breakpoint 768px changed to 643px;
--nav-fs variable on links; old "left:" offsets removed.

```
header {
  z-index: 2000; /* Task 1: mega-menu must sit above the grid, hero and any sticky element */
  position: relative;

}

#header-bottom {
  position: relative; /* Task 1: containing block for .mega-panel */
  display: flex;
  background-color: transparent;
  padding: 1.2em;
  justify-content: center;
  border-top: 0.1em;
  border-top-style: solid;
  border-color: black;
}

@media(min-width: 643px) {
  #header-bottom:hover {
    background-color: rgb(227, 6, 19);
    border-top: 0.1em;
    border-top-style: solid;
    border-color: white;
  }
}

.nav-link {
  display: inline;
  text-decoration: none;
  color: white;
  font-family: "Michroma", sans-serif;
  font-size: var(--nav-fs, 0.9em); /* --nav-fs is set per width in the Task 1 section */
  font-weight: bolder;
  padding-left: 0.5em;
  padding-right: 0.5em;
  position: relative;
```

```
}

#header-bottom-product {
  position: relative; /* Task 1: containing block for .mega-panel */
  display: flex;
  background-color: transparent;
  padding: 1.2em;
  justify-content: center;
  border-top: 0.1em;
  border-top-style: solid;
  border-color: white;
  background-color: rgb(227, 6, 19);
}

@media(min-width: 643px) {
  #header-bottom-product:hover {
    background-color: rgb(227, 6, 19);
    border-top: 0.1em;
    border-top-style: solid;
    border-color: white;
  }
}

style.css – Task 1 section (lines 1203–1431)
/*
================================================================================
   TASK 1 - NAVBAR + "SHOP BY CATEGORY" MEGA-MENU  (HTML + CSS only)
   N = 43  ->  BP = 600 + 43 = 643px   |   K = (43 mod 3) + 3 = 4 columns
   Three header layouts:
     >= 1100px      one row (original look)
     643 - 1099px   two rows: logo + icons, then the links
     <= 642px       hamburger menu, mega-menu becomes a <details> accordion

================================================================================
*/

/* ---------- shared defaults ---------- */
.menu-toggle { display: none; }          /* only used below BP */
.nav-cart { display: flex; align-items: center; }   /* Task 2: cart icon links to checkout.html */
.nav-menu { display: contents; }         /* links stay direct flex items of <nav>, as before */
```

```
.mega-accordion { display: none; }       /* only used below BP */

/* ---------- BP and wider: full-width panel ---------- */
@media (min-width: 643px) {
  .nav-link-stack { text-align: center; }

  /* The item stretches the full height of the bar and 1.2rem further down, so the
     pointer never leaves it on the way from the link to the panel. */
  .mega-item {
    align-self: stretch;
    display: flex;
    align-items: center;
    padding-bottom: var(--bar-pad);
    margin-bottom: calc(-1 * var(--bar-pad));
    /* NOT positioned on purpose: the panel is positioned against #header-bottom */
  }

  .mega-panel {
    --mega-cols: 4;                    /* K */
    position: absolute;                /* out of flow: never pushes the page down */
    top: 100%;                         /* directly under the navbar */
    left: 0;
    right: 0;                          /* full width */
    display: grid;
    grid-template-columns: repeat(var(--mega-cols), minmax(0, 1fr));
    gap: 1.5rem 2rem;
    padding: 1.75rem max(1rem, calc((100% - 1220px) / 2)) 2rem;
    background: #fff;
    color: #111;
    border-top: 3px solid rgb(147, 4, 12);
    box-shadow: 0 14px 28px rgba(0, 0, 0, 0.3);
    font-family: system-ui, -apple-system, BlinkMacSystemFont, 'Segoe UI', Roboto, Oxygen,
Ubuntu, Cantarell, 'Open Sans', 'Helvetica Neue', sans-serif;
    text-align: left;
    /* closed: invisible AND not focusable, so Tab skips it */
    visibility: hidden;
    opacity: 0;
    transform: translateY(-4px);
    /* closing is delayed by 0.25s ("hover intent") so a diagonal pointer path that briefly
       touches a neighbouring link on the way to the panel does not close it */
```

```
    transition: opacity 0.12s ease 0.25s, transform 0.12s ease 0.25s, visibility 0s linear 0.37s;
  }

  /* open on hover of the item (link OR panel) and on keyboard focus anywhere inside it */
  .mega-item:hover .mega-panel,
  .mega-item:focus-within .mega-panel {
    visibility: visible;
    opacity: 1;
    transform: none;
    transition-delay: 0s;              /* opening is instant */
  }

  .mega-heading {
    margin: 0 0 0.75rem;
    padding-bottom: 0.6rem;
    border-bottom: 1px solid #d9d9d9;
    font-family: "Michroma", sans-serif;
    font-size: 0.75rem;
    font-weight: 600;
    line-height: 1.3;
    letter-spacing: 0.04em;
    text-transform: uppercase;
    color: rgb(147, 4, 12);
  }

  .mega-panel ul { list-style: none; margin: 0; padding: 0; }
}

@media (min-width: 643px) and (prefers-reduced-motion: reduce) {
  .mega-panel { transition: none; transform: none; }
}

/* shared link look (panel + accordion) */
.mega-link {
  display: block;
  padding: 0.3rem 0;
  font-size: 0.9rem;
  line-height: 1.3;
  color: #111;
  text-decoration: none;
```

```
}
.mega-link:hover,
.mega-link:focus-visible {
  color: rgb(227, 6, 19);
  text-decoration: underline;
}

/* ---------- >= 1100px: single row ---------- */
@media (min-width: 1100px) {
  :root { --nav-fs: clamp(0.5rem, 1.1vw, 0.9rem); --bar-pad: 1.2rem; }
  #arsenal-cannon { margin-right: 2.5rem; }
}

/* ---------- 643px - 1099px: two rows (nine Michroma links do not fit in one) ---------- */
@media (min-width: 643px) and (max-width: 1099px) {
  :root { --nav-fs: clamp(0.5rem, calc((100vw - 3rem) / 64), 0.9rem); --bar-pad: 0.6rem; }

  #header-bottom,
  #header-bottom-product {
    flex-wrap: wrap;
    align-items: center;
    row-gap: 0.4rem;
    padding: 0.6rem 1rem;
  }
  nav { display: contents; }                    /* logo, icons and links share the bar */
  #arsenal-cannon { width: 3.5em; margin-right: auto; }
  .nav-icons,
  .nav-cart { order: 2; }                  /* row 1, right */
  .nav-menu {                                 /* row 2: the links */
    order: 3;
    flex: 0 0 100%;
    display: flex;
    justify-content: space-between;
    align-items: center;
  }
}

/* ---------- below BP (<= 642px): hamburger menu + accordion ---------- */
@media (max-width: 642px) {
  #hero { margin-top: -62px; }
```

```
#header-bottom,
#header-bottom-product {
  position: relative;
  width: 100%;
  height: 62px;
  padding: 0 1rem;
  justify-content: center;
  align-items: center;
}

nav { display: contents; }

#hamburger { display: flex; margin-right: 14px; }
#hamburger svg { width: 26px; height: 26px; }

.nav-icons { order: 3; left: 0; width: 24px; margin: 0 0 0 14px; }
.nav-icons:nth-of-type(2) { order: 2; margin: 0 auto 0 0; }
.nav-cart { order: 3; }

#arsenal-cannon {
  position: absolute;
  left: 50%;
  top: 50%;
  transform: translate(-50%, -50%);
  width: 5em;
}

/* the checkbox is visually hidden but still reachable with Tab / Space */
.menu-toggle {
  display: block;
  position: absolute;
  width: 1px;
  height: 1px;
  margin: 0;
  opacity: 0;
}
.menu-toggle:focus-visible + #hamburger { outline: 2px solid #fff; outline-offset: 4px; }

/* the collapsed menu: closed by default, opened by the checked checkbox */
```

```
.nav-menu {
  display: none;
  position: absolute;
  top: 100%;
  left: 0;
  right: 0;
  flex-direction: column;
  max-height: calc(100vh - 110px);
  overflow-y: auto;
  background: rgb(3, 3, 37);
  border-top: 1px solid rgba(255, 255, 255, 0.2);
}
.menu-toggle:checked ~ nav .nav-menu { display: flex; }

.nav-link,
.mega-acc-title {
  display: block;
  padding: 14px 16px;
  border-bottom: 1px solid rgba(255, 255, 255, 0.15);
  font-size: 0.8rem;
  text-align: left;
}
.nav-link br { display: none; }

/* "Shop by Category": the K groups as independent <details> */
.mega-trigger,
.mega-panel { display: none; }
.mega-accordion { display: block; }
.mega-acc-title {
  margin: 0;
  font-family: "Michroma", sans-serif;
  font-weight: bolder;
  color: #fff;
}
.mega-accordion details { border-bottom: 1px solid rgba(255, 255, 255, 0.15); }
.mega-accordion summary {
  display: flex;
  justify-content: space-between;
  align-items: center;
  padding: 12px 16px 12px 28px;
```

```
    list-style: none;
    cursor: pointer;
    color: #fff;
    font-family: system-ui, -apple-system, BlinkMacSystemFont, 'Segoe UI', Roboto, sans-serif;
    font-size: 0.95rem;
    font-weight: 600;
  }
  .mega-accordion summary::-webkit-details-marker { display: none; }
  .mega-accordion summary::after { content: "+"; font-size: 1.3rem; line-height: 1; }
  .mega-accordion details[open] > summary::after { content: "\2212"; }
  .mega-accordion ul { list-style: none; margin: 0; padding: 0 16px 12px 28px; }
  .mega-accordion .mega-link { color: rgba(255, 255, 255, 0.85); padding: 0.45rem 0; }
  .mega-accordion .mega-link:hover,
  .mega-accordion .mega-link:focus-visible { color: #fff; }
}
```

Task 2 – Checkout form (HTML5 validation only)
product.html and checkout.html – cart icon links to the checkout page (line 145)

```
      <a href="checkout.html" class="nav-cart" aria-label="Cart"><img class="nav-icons"
src="images/cart.svg" alt=""></a>
```

checkout.html – page content (lines 156–220)
The header and footer of checkout.html are the same as product.html; only <main> is new.

```
  <main>
    <section class="checkout">
      <h1 id="checkout-title">Checkout</h1>

      <!-- HTML5 validation only: required, type, pattern, minlength, min/max. No JavaScript on
this page. -->
      <form class="checkout-form" action="checkout.html" method="get">
        <div class="checkout-sets">

          <fieldset>
            <legend>Contact</legend>
            <div class="fields">
            <div class="field">
              <label for="fullname">Full name</label>
```

```
          <input type="text" id="fullname" name="fullname" required minlength="3"
maxlength="60" autocomplete="name" placeholder="e.g. Ali Khan"
aria-describedby="fullname-error">
          <span class="error" id="fullname-error">Enter your full name (at least 3
characters).</span>
        </div>
        <div class="field">
          <label for="email">Email</label>
          <input type="email" id="email" name="email" required autocomplete="email"
placeholder="name@example.com" aria-describedby="email-error">
          <span class="error" id="email-error">Enter a valid email address.</span>
        </div>
        <div class="field">
          <label for="phone">Phone</label>
          <input type="tel" id="phone" name="phone" required pattern="03[0-9]{2}-[0-9]{7}"
maxlength="12" inputmode="tel" autocomplete="tel" placeholder="03XX-XXXXXXX"
title="Format: 03XX-XXXXXXX" aria-describedby="phone-error">
          <span class="error" id="phone-error">Use the format 03XX-XXXXXXX.</span>
        </div>
        </div>
      </fieldset>

      <fieldset>
        <legend>Delivery</legend>
        <div class="fields">
        <div class="field">
          <label for="address">Address</label>
          <input type="text" id="address" name="address" required minlength="8"
maxlength="120" autocomplete="street-address" placeholder="House, street, area"
aria-describedby="address-error">
          <span class="error" id="address-error">Enter your full address (at least 8
characters).</span>
        </div>
        <div class="field">
          <label for="city">City</label>
          <select id="city" name="city" required aria-describedby="city-error">
            <option value="">Select a city</option>
            <option value="Lahore">Lahore</option>
            <option value="Karachi">Karachi</option>
            <option value="Islamabad">Islamabad</option>
```

```
            <option value="Rawalpindi">Rawalpindi</option>
            <option value="Faisalabad">Faisalabad</option>
            <option value="Multan">Multan</option>
            <option value="Peshawar">Peshawar</option>
            <option value="Quetta">Quetta</option>
          </select>
          <span class="error" id="city-error">Please choose a city.</span>
        </div>
        <div class="field">
          <label for="delivery-date">Delivery date</label>
          <input type="date" id="delivery-date" name="delivery_date" required
min="2026-10-29" max="2026-12-11" aria-describedby="delivery-date-error">
          <span class="error" id="delivery-date-error">Choose a date from 29 Oct to 11 Dec
2026.</span>
        </div>
        </div>
      </fieldset>

    </div>
    <button type="submit" class="checkout-submit">PLACE ORDER</button>
   </form>
  </section>
 </main>
```

style.css – Task 2 section

```
/*
================================================================================
  TASK 2 - CHECKOUT FORM (HTML5 validation only, no JavaScript)
  BP = 643px  |  H = 301  ->  accent colour hsl(301, 70%, 45%)

================================================================================
*/
.checkout {
 --accent: hsl(301, 70%, 45%);        /* H = (43 x 7) mod 360 = 301 */
 --error: #d60000;
 max-width: 1000px;
 margin: 0 auto 3rem;
 padding: 0 1rem;
 font-family: system-ui, -apple-system, BlinkMacSystemFont, 'Segoe UI', Roboto, Oxygen,
Ubuntu, Cantarell, 'Open Sans', 'Helvetica Neue', sans-serif;
```

```
}

#checkout-title {
  margin: 0 0 1.5rem;
  font-family: "Michroma", sans-serif;
  font-size: 1.4rem;
  font-weight: 600;
  color: black;
}

/* default (below BP): everything in one column, labels above inputs */
.checkout-sets { display: grid; grid-template-columns: 1fr; gap: 1.25rem; }

.checkout fieldset {
  min-width: 0;
  margin: 0;
  padding: 0.5rem clamp(0.75rem, 2.2vw, 1.25rem) 0.5rem;   /* tighter near BP so the inputs stay
roomy */
  border: 1px solid rgb(213, 210, 210);
  border-radius: 8px;
}
.checkout legend {
  padding: 0 0.5rem;
  font-family: "Michroma", sans-serif;
  font-size: 0.8rem;
  font-weight: 600;
  letter-spacing: 0.04em;
  color: var(--accent);
}

.field { margin-bottom: 1rem; }
.field label {
  display: block;
  margin-bottom: 0.3rem;
  font-size: 0.85rem;
  font-weight: 600;
  line-height: 1.3;
}
.field input,
.field select {
```

```
  display: block;
  width: 100%;
  padding: 0.6rem 0.75rem;
  border: 1px solid #b5b5b5;
  border-radius: 6px;
  background: #fff;
  font: inherit;
  font-size: 0.95rem;
  color: #111;
}
.field input:focus,
.field select:focus {
  outline: 2px solid var(--accent);
  outline-offset: 1px;
}

/* messages are hidden until the user has interacted with the field */
.error {
  display: none;
  margin-top: 0.3rem;
  font-size: 0.8rem;
  line-height: 1.3;
  color: var(--error);
}

/* :user-invalid only matches AFTER the user has typed/changed/left the field
   (or tried to submit), so nothing is red on page load */
.field input:user-invalid,
.field select:user-invalid {
  border-color: var(--error);
  border-width: 2px;
  padding: calc(0.6rem - 1px) calc(0.75rem - 1px);   /* keep the box the same size */
}
.field input:user-invalid + .error,
.field select:user-invalid + .error { display: block; }

/* older browsers without :user-invalid: text-like inputs that have been filled in */
@supports not selector(:user-invalid) {
  .field input:not(:placeholder-shown):invalid { border-color: var(--error); border-width: 2px; }
  .field input:not(:placeholder-shown):invalid + .error { display: block; }
```

```
}

/* submit button: full width at every size */
.checkout-submit {
  display: block;
  width: 100%;
  margin-top: 1.5rem;
  padding: 1rem;
  border: none;
  border-radius: 99px;
  background-color: rgb(147, 4, 12);
  color: white;
  font-family: "Michroma", sans-serif;
  font-size: 0.85rem;
  font-weight: 600;
  letter-spacing: 0.05em;
  cursor: pointer;
}
.checkout-submit:hover { background-color: rgb(227, 6, 19); }
.checkout-submit:focus-visible { outline: 3px solid var(--accent); outline-offset: 3px; }

/* BP and wider: fieldsets side by side, labels in a column to the left of the inputs */
@media (min-width: 643px) {
  .checkout-sets { grid-template-columns: 1fr 1fr; align-items: start; gap: 1.5rem; }

  .fields {
    display: grid;
    grid-template-columns: clamp(4.5rem, 11vw, 6rem) minmax(0, 1fr);   /* label column | input
column */
    column-gap: clamp(0.6rem, 1.6vw, 1rem);
    row-gap: 1rem;
    align-items: start;
    padding: 0.5rem 0;
  }
  .field { display: contents; }                  /* label, input and message become grid items */
  .field label { margin: 0; padding-top: 0.7rem; }
  .field .error { grid-column: 2; margin: -0.5rem 0 0; }
}
```

Task 3 – JavaScript: cart badge and mini cart
script.js (whole file)

```
// Task 3 - cart badge and mini cart (plain JavaScript, no libraries)

// N = 43  ->  a product may be added at most (N mod 4) + 2 = (43 mod 4) + 2 = 5 times
const N = 43;
const MAX_PER_PRODUCT = (N % 4) + 2;

// The cart is an array of { name, price } objects.
// Adding the same product twice puts two entries in the array.
const cart = [];

// ----- elements already on the page -----
const addButtons = document.querySelectorAll('.add-eqt');          // the "+" (Add to Cart) buttons
const cartLink   = document.getElementById('cart-toggle');         // the cart icon
const badge      = document.getElementById('cart-badge');
const panel      = document.getElementById('mini-cart');
const listEl     = document.getElementById('mini-cart-list');
const emptyEl    = document.getElementById('mini-cart-empty');
const totalEl    = document.getElementById('mini-cart-total');

// ----- helpers -----

// "£70.00" -> 70
function parsePrice(text) {
  return Number(text.replace(/[^0-9.]/g, ''));
}

// 70 -> "£70.00"
function formatPrice(value) {
  return '£' + value.toFixed(2);
}

// Read name and price from the card that contains the clicked button.
function readProduct(button) {
  const card = button.closest('.EQT-card, .product-product-card');
  return {
    name: card.querySelector('.eqt-title, .product-title').textContent.trim(),
    price: parsePrice(card.querySelector('.eqt-price, .product-price').textContent)
  };
```

```
}

// How many of this product are in the cart right now?
function countInCart(name) {
  return cart.filter(function (item) { return item.name === name; }).length;
}

// ----- add to cart -----
function addToCart(button) {
  const product = readProduct(button);
  if (countInCart(product.name) >= MAX_PER_PRODUCT) return;   // safety net, button is
disabled anyway
  cart.push(product);
  render();
}

// ----- remove from cart -----
function removeFromCart(index) {
  cart.splice(index, 1);
  render();
}

// ----- keep every "+" button in sync with the cart -----
function updateButtons() {
  addButtons.forEach(function (button) {
    const name = readProduct(button).name;
    if (countInCart(name) >= MAX_PER_PRODUCT) {
      button.textContent = 'Limit reached';
      button.disabled = true;
      button.classList.add('is-limit');
      button.setAttribute('aria-label', 'Limit reached for ' + name);
    } else {
      button.textContent = '+';
      button.disabled = false;
      button.classList.remove('is-limit');
      button.setAttribute('aria-label', 'Add ' + name + ' to cart');
    }
  });
}
```

```
// ----- redraw badge, list, total and buttons from the cart array -----
function render() {
  // list
  listEl.replaceChildren();
  cart.forEach(function (item, index) {
    const li = document.createElement('li');
    li.className = 'mini-cart-item';

    const name = document.createElement('span');
    name.className = 'mini-cart-name';
    name.textContent = item.name;

    const price = document.createElement('span');
    price.className = 'mini-cart-price';
    price.textContent = formatPrice(item.price);

    const remove = document.createElement('button');
    remove.type = 'button';
    remove.className = 'mini-cart-remove';
    remove.textContent = 'Remove';
    remove.dataset.index = index;
    remove.setAttribute('aria-label', 'Remove ' + item.name + ' from cart');

    li.append(name, price, remove);
    listEl.append(li);
  });
  emptyEl.hidden = cart.length > 0;

  // total, calculated with reduce()
  const total = cart.reduce(function (sum, item) { return sum + item.price; }, 0);
  totalEl.textContent = formatPrice(total);

  // badge on the cart icon
  badge.textContent = cart.length;
  badge.hidden = cart.length === 0;
  cartLink.setAttribute('aria-label', 'Cart, ' + cart.length + (cart.length === 1 ? ' item' : ' items'));

  updateButtons();
}
```

```
// ----- open / close the mini cart -----
function setOpen(open) {
  panel.hidden = !open;
  cartLink.setAttribute('aria-expanded', String(open));
}

// ----- events -----
addButtons.forEach(function (button) {
  button.addEventListener('click', function () { addToCart(button); });
});

// Clicking the cart icon toggles the panel instead of following the link
// (the panel has its own CHECKOUT link, and with no JavaScript the icon still goes to
checkout.html).
cartLink.addEventListener('click', function (event) {
  event.preventDefault();
  setOpen(panel.hidden);
});

// One listener on the list handles every Remove button (event delegation).
listEl.addEventListener('click', function (event) {
  const button = event.target.closest('.mini-cart-remove');
  if (!button) return;
  const index = Number(button.dataset.index);
  removeFromCart(index);

  // keep keyboard focus inside the panel after the row disappears
  const next = listEl.querySelectorAll('.mini-cart-remove')[index] ||
               listEl.querySelectorAll('.mini-cart-remove')[index - 1];
  (next || cartLink).focus();
});

// Escape closes the panel and returns focus to the cart icon
document.addEventListener('keydown', function (event) {
  if (event.key === 'Escape' && !panel.hidden) {
    setOpen(false);
    cartLink.focus();
  }
});
```

```
// first draw (sets button labels, hides the badge)
render();
```

index.html – cart icon with badge and mini-cart panel (lines 147–160)

```
      <!-- Task 3: cart icon = badge + mini cart. Without JavaScript the link still goes to
checkout.html -->
      <div class="nav-cart">
        <a href="checkout.html" class="cart-link" id="cart-toggle" aria-label="Cart, 0 items"
aria-haspopup="true" aria-expanded="false" aria-controls="mini-cart">
          <img class="nav-icons" src="images/cart.svg" alt="">
          <span class="cart-badge" id="cart-badge" hidden>0</span>
        </a>
        <div class="mini-cart" id="mini-cart" hidden>
          <h2 class="mini-cart-title">Your cart</h2>
          <p class="mini-cart-empty" id="mini-cart-empty">Your cart is empty.</p>
          <ul class="mini-cart-list" id="mini-cart-list"></ul>
          <p class="mini-cart-total">Total: <strong id="mini-cart-total">£0.00</strong></p>
          <a href="checkout.html" class="mini-cart-checkout">CHECKOUT</a>
        </div>
      </div>
```

index.html – script link (end of body) and the "+" buttons (type="button" added)

```
<script src="script.js"></script>

      <button class="add-eqt" type="button">+</button>
```

style.css – Task 3 section

```
/*
================================================================================
  Task 3 - cart badge, mini cart, "Limit reached" button
  Limit = (N mod 4) + 2 = 5 (set in script.js)

================================================================================
*/

/* the wrapper is the positioning context for the badge and the panel */
.nav-cart { position: relative; }
.cart-link { position: relative; display: flex; align-items: center; }

.cart-badge {
```

```
  position: absolute;
  top: -9px;
  right: -6px;
  min-width: 18px;
  height: 18px;
  padding: 0 4px;
  box-sizing: border-box;
  border: 2px solid #fff;              /* readable on the dark bar and on the red hover bar */
  border-radius: 9px;
  background: rgb(227, 6, 19);
  color: #fff;
  font: 700 10px/14px system-ui, -apple-system, 'Segoe UI', Roboto, sans-serif;
  text-align: center;
  pointer-events: none;                /* clicks go to the icon underneath */
}
.cart-badge[hidden] { display: none; }

/* mini-cart panel: out of flow, directly under the cart icon, above everything (header z-index is
2000) */
.mini-cart {
  position: absolute;
  top: calc(100% + 0.9rem);
  right: 0;
  width: 320px;
  max-width: calc(100vw - 2rem);
  box-sizing: border-box;
  padding: 1rem;
  background: #fff;
  color: #111;
  border-top: 3px solid rgb(147, 4, 12);
  box-shadow: 0 14px 28px rgba(0, 0, 0, 0.3);
  font-family: system-ui, -apple-system, BlinkMacSystemFont, 'Segoe UI', Roboto, Oxygen,
Ubuntu, Cantarell, 'Open Sans', 'Helvetica Neue', sans-serif;
  font-size: 0.9rem;
  text-align: left;
  z-index: 10;
}
.mini-cart[hidden] { display: none; }

.mini-cart-title {
```

```
  margin: 0 0 0.75rem;
  padding-bottom: 0.6rem;
  border-bottom: 1px solid #d9d9d9;
  font-family: "Michroma", sans-serif;
  font-size: 0.75rem;
  font-weight: 600;
  letter-spacing: 0.04em;
  text-transform: uppercase;
  color: rgb(147, 4, 12);
}

.mini-cart-empty { margin: 0 0 0.75rem; color: #555; }
.mini-cart-empty[hidden] { display: none; }

.mini-cart-list {
  list-style: none;
  margin: 0;
  padding: 0;
  max-height: 260px;
  overflow-y: auto;
}

.mini-cart-item {
  display: flex;
  align-items: center;
  gap: 0.6rem;
  padding: 0.5rem 0;
  border-bottom: 1px solid #eee;
}
.mini-cart-name  { flex: 1; min-width: 0; line-height: 1.3; }
.mini-cart-price { font-weight: 700; white-space: nowrap; }

.mini-cart-remove {
  padding: 0.25rem 0.5rem;
  border: 1px solid rgb(147, 4, 12);
  border-radius: 6px;
  background: #fff;
  color: rgb(147, 4, 12);
  font-size: 0.75rem;
  font-weight: 700;
```

```
  cursor: pointer;
}
.mini-cart-remove:hover,
.mini-cart-remove:focus-visible { background: rgb(147, 4, 12); color: #fff; }

.mini-cart-total {
  display: flex;
  justify-content: space-between;
  margin: 0.75rem 0;
  font-size: 1rem;
}

.mini-cart-checkout {
  display: block;
  padding: 0.6rem;
  border-radius: 6px;
  background: rgb(147, 4, 12);
  color: #fff;
  font-weight: 700;
  text-align: center;
  text-decoration: none;
}
.mini-cart-checkout:hover,
.mini-cart-checkout:focus-visible { background: rgb(227, 6, 19); }

/* "Limit reached" state of the + button: wider pill, right-aligned on the card, greyed out */
.add-eqt.is-limit {
  display: block;
  width: auto;
  height: 26px;
  margin-left: auto;
  padding: 0 8px;
  font-size: 11px;
  font-weight: 700;
  white-space: nowrap;
  transform: translate(-8px, -40px);
}
.add-eqt:disabled {
  background-color: #777;
  color: #fff;
```

```
  cursor: not-allowed;
}
```